

NIKKO, Japan

WMO: 476900

Lat: 36.7397N

Lon: 139.5003E

Elev: 4245

StdP: 12.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	14.5	-0.5	6.2	20.2	1.9	7.1	20.0	32.0	26.7	29.2	26.7	8.8	260	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	77.2	68.6	75.1	68.0	73.0	66.9	70.7	74.8	69.3	73.1	68.1	71.4	6.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	126.2	73.2	68.1	121.1	71.8	67.0	116.3	70.3	37.6	75.1	36.4	73.4	35.2	71.3	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	22.2	18.8	9.2	81.5	2.0	1.9	7.8	82.8	6.6	83.9	5.4	85.0	4.0	86.3		
			DB	WB	7.4	72.6	1.9	1.1	6.1	73.4	5.0	74.0	3.9	74.6	2.6	75.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	24.4	25.4	31.3	41.2	50.7	57.2	65.2	66.5	59.6	49.5	39.7	30.1
	DBStd	15.59	5.28	6.17	6.57	6.61	5.04	4.80	4.40	4.00	5.63	5.67	5.95	6.09
	HDD50	3398	793	690	580	272	51	3	0	0	3	78	311	617
	HDD65	7386	1258	1110	1045	715	442	238	53	31	174	480	758	1082
	CDD50	1641	0	0	0	7	74	220	471	511	292	63	3	0
	CDD65	154	0	0	0	0	0	4	60	77	13	0	0	0
	CDH74	300	0	0	0	0	1	5	125	155	14	0	0	0
	CDH80	8	0	0	0	0	0	0	3	5	0	0	0	0
Wind	WSAvg	7.1	9.9	9.5	8.9	7.7	6.5	5.1	4.9	4.9	5.1	5.9	7.8	9.5
Precipitation	PrecAvg	72.1	1.9	1.9	3.6	5.4	6.5	7.7	9.6	10.3	12.1	7.2	3.9	2.1
	PrecMax	91.9	6.3	5.2	7.4	10.1	15.4	22.6	17.1	22.2	26.2	20.6	12.7	6.9
	PrecMin	44.7	0.3	0.4	0.8	1.2	2.5	2.6	4.3	3.2	1.3	0.5	0.1	0.2
	PrecStd	10.9	1.4	1.2	1.5	2.7	2.8	3.8	3.6	5.3	6.3	4.4	2.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.3	48.7	54.6	65.4	71.1	74.3	79.8	80.5	76.0	68.0	58.6	51.2
		MCWB	38.7	41.1	43.0	50.8	57.2	64.1	68.1	69.0	66.2	59.5	48.9	45.2
	2%	DB	39.5	43.4	49.8	59.4	66.9	70.6	77.2	77.7	72.7	64.3	54.8	45.9
		MCWB	34.0	36.6	40.8	48.2	54.6	62.4	68.7	69.1	65.7	58.7	48.0	40.5
	5%	DB	36.4	38.8	45.8	55.9	63.6	68.0	75.1	75.5	70.1	61.2	52.0	42.8
		MCWB	31.2	33.3	39.0	46.8	53.7	61.3	68.1	68.8	64.7	56.3	46.7	37.3
	10%	DB	33.4	35.3	42.3	52.6	60.8	65.4	72.8	73.5	67.6	58.4	49.5	39.7
		MCWB	28.7	30.6	36.5	44.5	52.5	60.3	66.9	67.9	63.7	54.4	45.2	35.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	43.2	46.7	55.9	61.6	67.3	72.3	72.4	69.2	62.8	53.3	47.7
		MCDB	42.7	47.4	50.6	58.8	65.9	70.7	76.6	76.9	72.6	65.0	55.0	50.0
	2%	WB	35.4	38.1	43.2	51.9	58.5	64.8	70.6	71.0	67.2	60.1	50.6	41.9
		MCDB	38.7	42.0	47.4	56.8	62.6	68.2	74.9	75.2	70.3	62.8	52.7	44.6
	5%	WB	31.6	34.2	40.3	48.9	56.3	63.2	69.3	69.8	65.9	57.7	48.0	38.4
		MCDB	35.3	37.7	44.5	53.1	60.8	66.3	73.2	73.6	68.5	60.1	50.6	41.5
	10%	WB	29.0	31.3	37.3	46.4	54.2	61.4	67.8	68.6	64.5	55.2	45.7	35.4
		MCDB	33.0	34.7	41.6	50.5	58.7	64.2	71.1	72.1	67.2	57.5	48.6	38.9
Mean Daily Temperature Range	5% DB	MDBR	12.1	13.4	14.2	15.8	15.6	12.0	11.1	11.4	11.1	12.6	13.2	12.1
		MCDBR	15.2	17.6	18.7	21.4	20.7	16.0	14.9	15.0	13.6	15.1	16.0	15.7
		MCWBR	12.2	13.9	13.9	14.4	13.5	10.2	9.1	9.4	8.7	10.8	12.4	12.3
	5% WB	MCDBR	14.5	16.3	17.4	18.3	17.5	12.8	13.6	13.8	11.8	12.6	14.3	14.8
		MCWBR	12.7	14.0	14.7	14.0	12.7	8.9	8.9	9.1	8.1	9.8	12.4	13.1
Clear-Sky Solar Irradiance	taub	0.303	0.337	0.394	0.437	0.459	0.489	0.493	0.492	0.423	0.379	0.334	0.307	
	taud	2.294	2.182	2.086	2.050	2.035	2.008	2.071	2.068	2.234	2.300	2.382	2.376	
	Ebn at Noon	279	281	274	268	264	255	252	250	262	264	266	271	
	Edh at Noon	31	39	48	52	54	56	52	51	41	35	29	27	
All-Sky Solar Radiation	RadAvg	742	975	1258	1482	1631	1483	1448	1455	1180	957	753	658	
	RadStd	47	70	79	129	166	113	224	159	98	89	62	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page